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School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 06-05-2026

TICKET BOOKING SYSTEM FOR TECHNICAL FEST

SDG: SDG 9 – Industry, Innovation & Infrastructure

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- Most technical festivals still rely on manual registration or fragmented department-level spreadsheets.

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- This reliance leads to data silos, ticket overselling, and delayed communication for venue changes.

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- Our project is vital because it centralizes event management into a single, unified microservices platform.

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- It ensures a seamless user experience by automating ticket inventory and personal booking history.

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- Real-world relevance lies in providing instant, automated notifications for any administrative updates.

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- The system utilizes a distributed architecture to handle high traffic from multiple departments simultaneously.

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- The project comprises four specialized services: Discovery, User Auth, Event Inventory, and Booking.

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- In summary, it bridges the gap between organizers and participants with real-time, data-driven management.

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PROBLEM STATEMENT



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In many academic institutions, current systems for managing large-scale technical festivals are not fully automated and rely heavily on fragmented manual work across different departments. This leads to significant inefficiencies in managing attendee data, tracking live ticket inventory across multiple events, and ensuring accurate communication. Furthermore, existing systems lack a centralized platform for students to explore all available fests or receive real-time notifications regarding venue or schedule updates. Consequently, there is an urgent need for a simple, reliable, and automated microservices-based system that can improve coordination, accuracy, and the overall user experience during the technical fest registration process.

OBJECTIVE(S)



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- To design and develop a scalable microservices-based web portal that centralizes technical fest registration and automates ticket inventory management for multiple departments.
- To implement a real-time notification engine and a dedicated administrator dashboard to ensure live updates and transparent monitoring of global booking logs.

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PROPOSED SYSTEM



- 1 • **System Description:** A distributed microservices architecture utilizing Spring Boot and React to manage fests, users, and bookings as independent, modular services.
- 2 • **Unified Frontend:** Centralizes all flagship technical fests onto a single platform, allowing students to browse and register without switching between multiple department sites.
- 3 • **Dynamic Inventory Management:** Solves the issue of ticket overselling by using real-time availability counters that sync instantly with the global database during every transaction.
- 4 • **Multi-Attendee Automation:** Addresses complex registration needs by enabling a single booking to dynamically capture unique data (name, email, dept) for every individual ticket.
- 5 • **Broadcast Notification System:** Resolves communication delays by using a notification engine to send instant alerts to attendees regarding venue or schedule changes.
- 6 • **Administrative Transparency:** Provides a global booking log and master event control dashboard, giving organizers real-time visibility that current manual systems lack.
- 7 • **High-Availability Advantage:** By using the Eureka Discovery Server, the system ensures that even if one service undergoes maintenance, the rest of the portal remains operational.
- 8 • **Scalable & Premium UX:** Offers a modern, glassmorphic user interface that can handle a larger volume of traffic and data more efficiently than traditional spreadsheet-based methods.
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PROPOSED SYSTEM ARCHITECTURE



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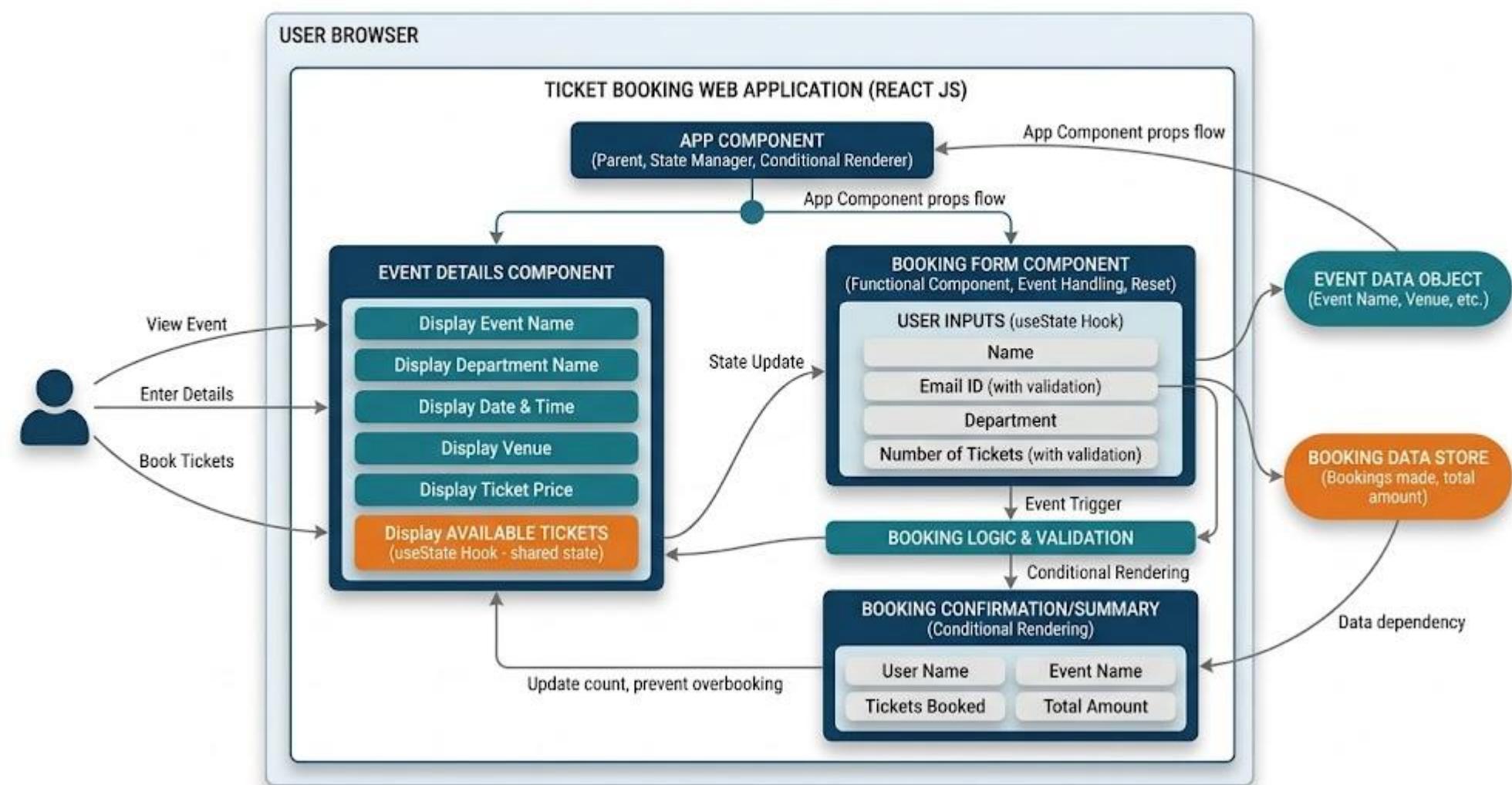
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Component	Description
Input	User login credentials, registration details (Name, Email, Department), event selection, and specific attendee data for each ticket booked (Multi-attendee input).
Processing	Role-based authentication (Spring Security), real-time ticket inventory calculation, cross-service communication via RestTemplate for booking validation, and automated notification triggering.
Database	Relational data storage using MySQL . It persists User accounts, Event catalogues (seeded data), live Booking transactions, and serialized Attendee lists.
Output	Dynamic Dashboards (Admin/User), successful booking confirmation (Interactive Ticket Popup), live notification alerts/badges, and history logs of all technical fest participations.

WORKFLOW



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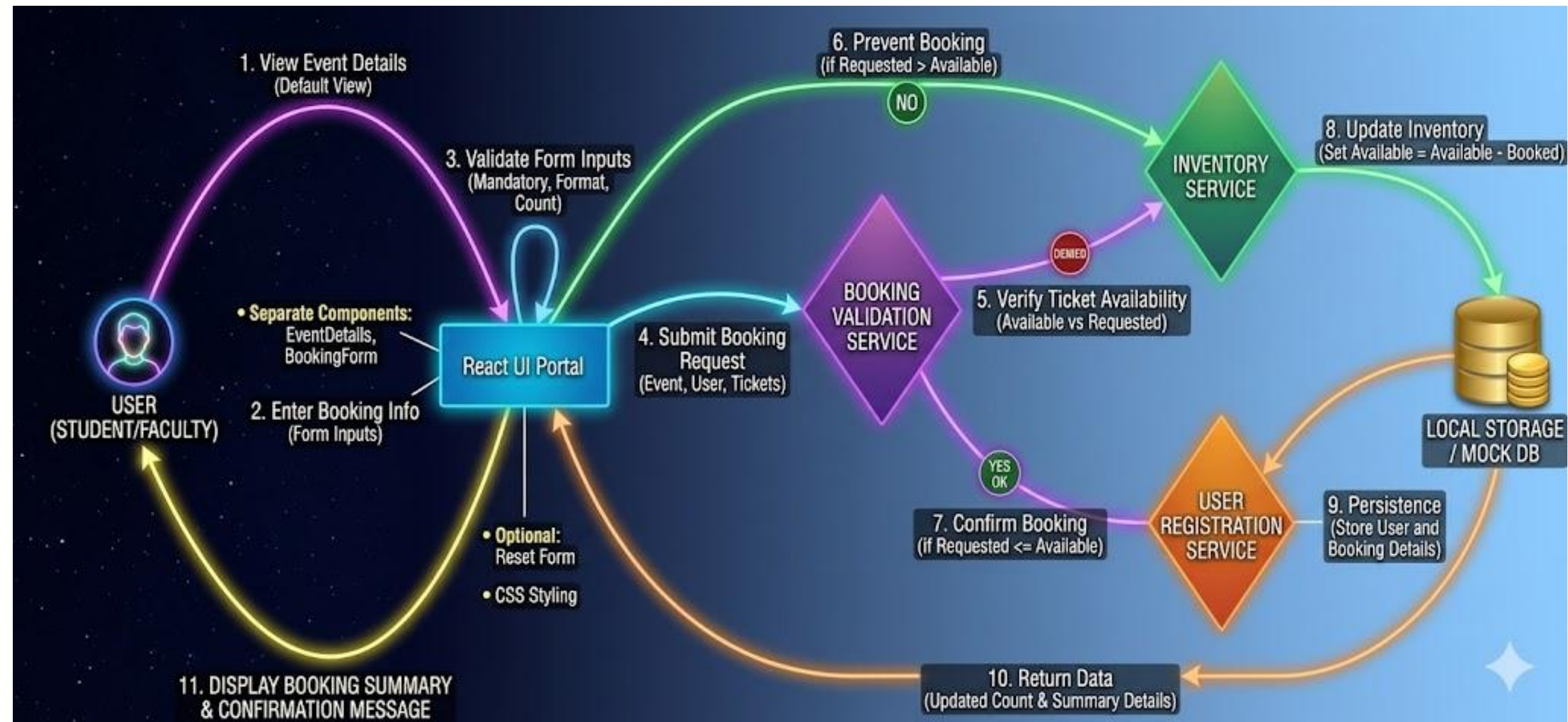
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TECHNOLOGIES USED



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- **Frontend:** React JS, CSS3, HTML5

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- **Backend:** Java 17+, Spring Boot, Spring Data JPA, Hibernate, Eureka Service Registry.

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- **Database:** MySQL

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- **Tools:** VS Code (IDE), Eclipse(Backend), Postman (API Testing)

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- **Hardware/Software Requirements:**

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OS: Windows 10/11

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RAM: 8GB

Processor: Intel i3/i5/ high processor

MODULE DESCRIPTION



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Module 1: Identity & Access Management (User Service) - Handles secure user registration, multi-role authentication (Admin/User), and maintains a persistent notification engine for real-time user alerts.

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Module 2: Festive Catalogue & Inventory (Event Service) - Manages the master directory of all 10+ technical fests, including department-specific details, live ticket availability, and dynamic pricing models.

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Module 3: Transactional Engine (Booking Service) - Processes complex multi-attendee ticket purchases, validates availability across services, and ensures data consistency during high-volume bookings.

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Module 4: Global Control & Reporting (Admin Dashboard) - Provides administrative tools for CRUD operations on events, visibility into the global booking log, and the ability to broadcast urgent notifications to participants.

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IMPLEMENTATION SCREENSHOTS



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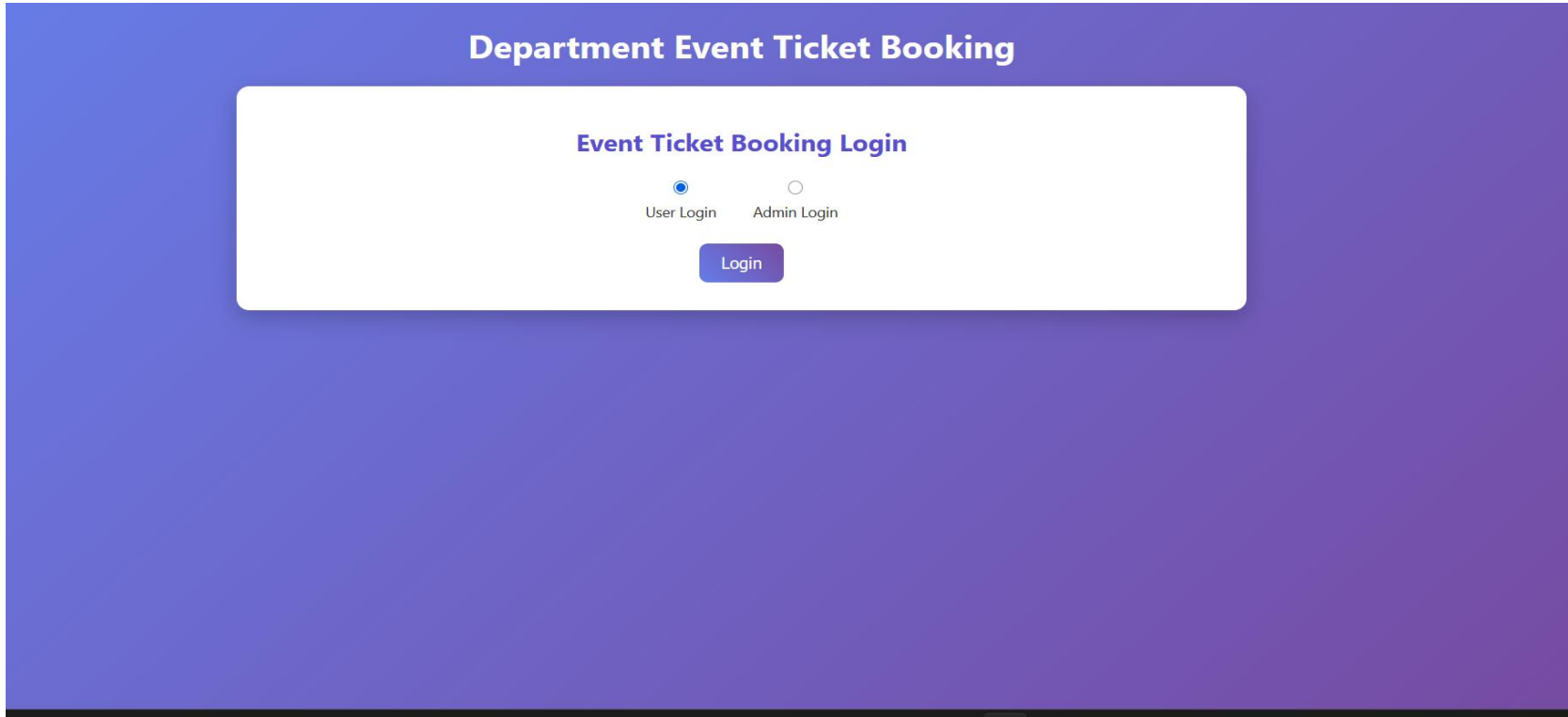
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Home Page – Displays about the website in short and leads to the next features

Events Page – Displays the list of the upcoming events which helps user to have an overview of events happening



IMPLEMENTATION SCREENSHOTS



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Book Tickets

Enter your name

Enter your email

Enter your department

Enter number of tickets

Book Now Reset

Register Page – It helps the users/ admin to create their account

Booking Summary

Payment successful and booking confirmed!

Booking ID: 1778018868607

User Name: Pavani

Email: vtu25989@veltech.edu.in

Department: CSE

Event Name: AI & ML Workshop

Tickets Booked: 4

Total Amount: ₹600

Payment Status: Paid

Download Invoice

IMPLEMENTATION SCREENSHOTS



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Login

☐ User Login

☒ Admin Login

Login

Admin Panel

Only admin can monitor bookings, revenue, and remaining tickets here.

Admin Dashboard

Total Bookings
1

Tickets Sold
5

Total Revenue
₹1,000.00

Remaining Tickets
45

Reset All Bookings

Booking Records

ID	Name	Email	Department	Tickets	Amount
1	Pavani	pavani207008@gmail.com	CSE	5	₹1,000.00

Admin Page – Here Admin can manage users, events, bookings, etc.

IMPLEMENTATION SCREENSHOTS



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User Page – Here user can explore about various events , book tickets and view them

Event Details

Event Name: AI & ML Workshop

Department: Department of AI & DS

Date: 15 June 2026

Time: 2:00 PM

Venue: Lab Complex

Ticket Price: ₹150

Available Tickets: 40

Book Tickets

Proceed to Payment

Reset

Booking Summary

Payment successful and booking confirmed!

Invoice No: INV-1777800307806

Payment ID: PAY-1777800321760

User Name: Pavani

Email: pavani207008@gmail.com

Department: CSE

Event Name: Department of Computer Science and Engineering Annual Technical Fest 2026 – Coding Contest, Project Expo and Seminar Series

Event Date: 10 June 2026

Event Time: 10:00 AM

Venue: Seminar Hall A

Tickets Booked: 5

Total Amount: ₹1,000.00

Payment Status: Paid

Download Invoice

ADVANTAGES & LIMITATIONS

Advantages



- 1 • **Distributed Architecture:** Enhances system fault tolerance through microservice decoupling, ensuring that failure in one service does not crash the entire portal.
- 2 • **Inventory Integrity:** Guarantees real-time ticket validation across concurrent users to prevent overselling through a dedicated event-logic controller.
- 3 • **Dynamic UX:** Features a live notification engine that provides immediate feedback and alerts regarding administrative updates directly on the user dashboard.
- 4 • **Rich Asset Handling:** Implements a unique, glassmorphic design system that ensures a premium and responsive interface for both desktop and mobile views.
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Limitations

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- 7 • **Payment Processing:** Currently focuses on the technical registration workflow and does not yet include a live integrated third-party payment gateway.
- 8 • **Physical Ticketry:** While it generates interactive digital logs, it lacks automated PDF generation and QR-code scanning for on-spot physical check-ins.
- 9 • **Cloud Persistence:** Designed for a local localized server environment, which would require Dockerization for horizontal scaling during massive traffic spikes.
- **Dependency Constraints:** Requires a stable network connection to the Eureka Discovery server and MySQL instance to process any transactional bookings.

Thank you

